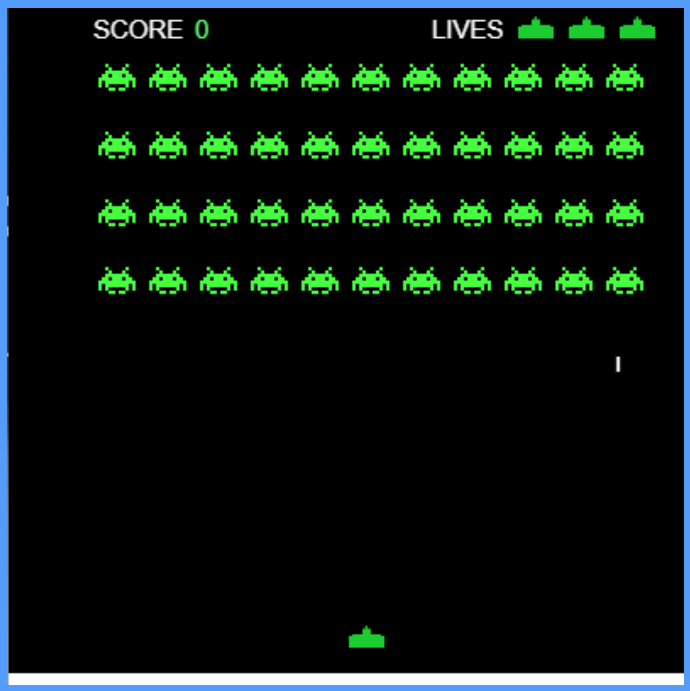
LITTLE SCIENTISTS ACADEMY

micro:bit Course

Module: Micro:bit Game Design

Arcade Games

# SPACE INVADERS




## Lesson 1

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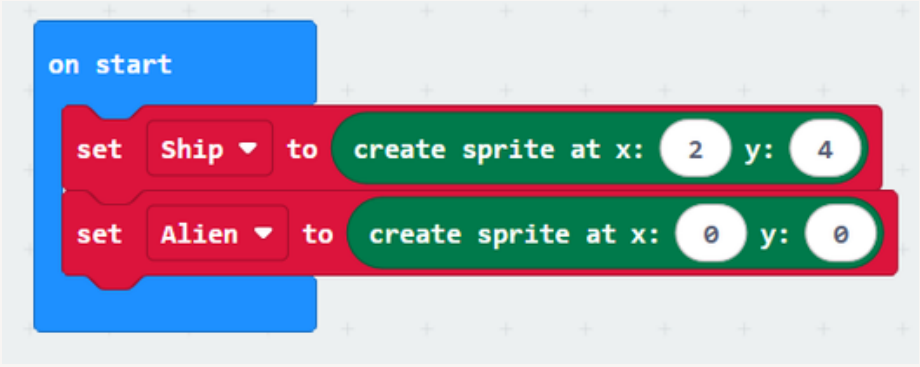
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# MICRO:BIT SPACE INVADERS

You are going to make Space Invaders on the micro:bit! You will code the A&B buttons to fire lasers at the alien ship.

1



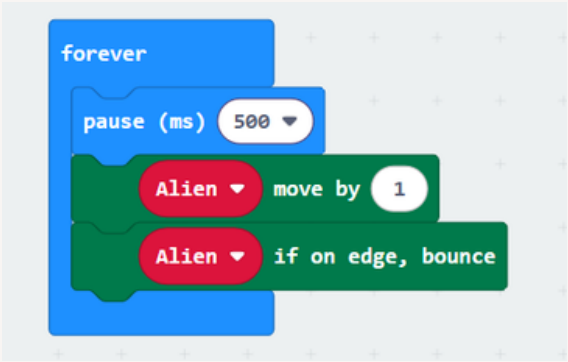

Add the alien and ship sprites to 'on start'

2

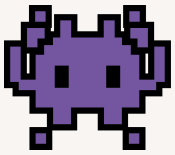



Use the input blocks to move the ship left or right

3




Add an infinite loop to make the alien spacecraft move from left to right.

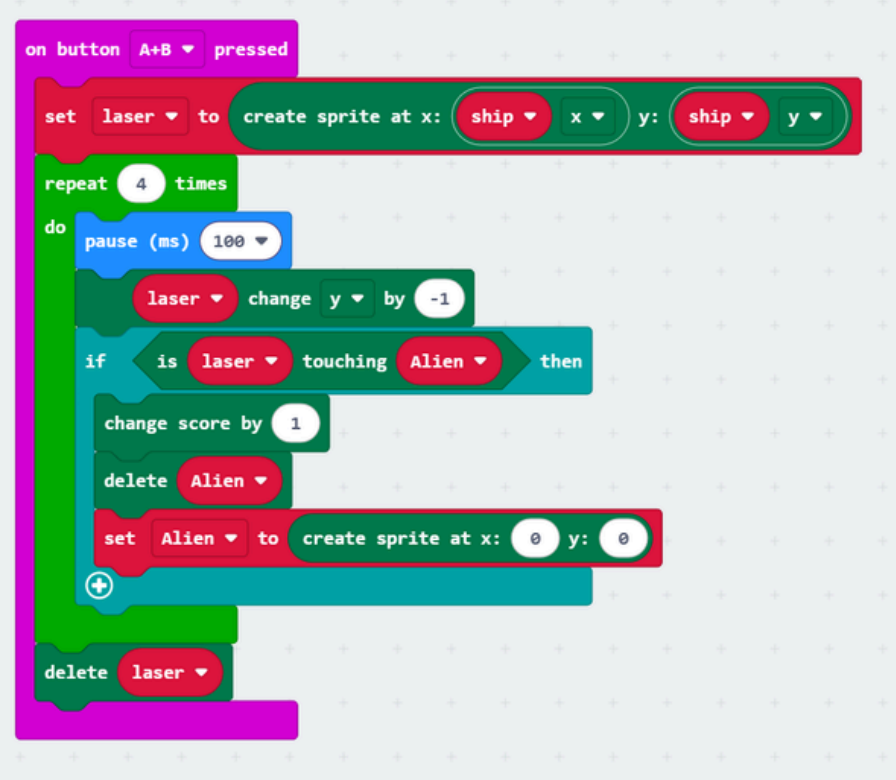


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# MICRO:BIT SPACE INVADERS

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4




This loop is a bit more complicated.
First of all, it uses the A+B input.
In our variables, we create a new, unprotected 'Laser' and position it exactly where our ship is.
We create a repeat loop set to 5 (which is the size of our screen (5×5 pixels)). After a very short pause, the laser moves to the top of the screen.
If the laser hits the alien, the alien regenerates part of its body and the score increases by 1.
The laser is destroyed at the top of the screen

## CHALLENGE

1) Alien Bomb Added: The alien ship now periodically drops a bomb downward toward your defense ship.

Player Defeat Condition: If the alien bomb hits your defense ship, the game ends (Game Over).

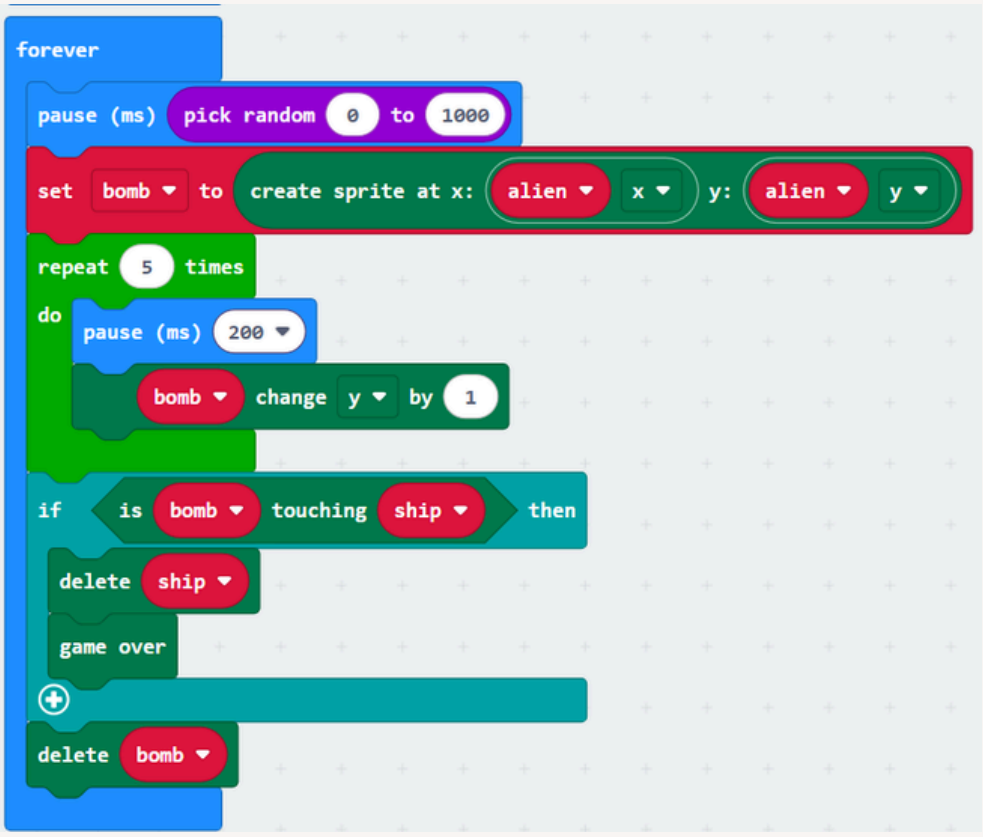
Dynamic Alien Speed: Every time your score increases, the alien ship's movement speed speeds up slightly.

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# Possible solution

1

Bomb




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